

A Punctured Lower-Even Boundary-Orbit Ladder

Bin Wang

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Abstract

After removing the selected orders $2k$ and $2k - 2$, the one-alias prefix still contains a complete ladder of physical boundary orbits at $n = 2m$, $m_\star \leq m \leq k - 2$. We extract all of them simultaneously. Writing

$$G_m = \frac{r_H^{-2m}}{1 + |M_m|}, \quad F_m^{\text{orb}} = 2mG_m,$$

the exact direct coefficient is

$$p_{\sigma,k,2m} = Y_{k,m} + \mathcal{P}_{\sigma,2m} - S_{k,m}, \quad S_{k,m} = F_m^{\text{orb}} + \mathcal{A}_{k,2m}.$$

Let $x = (\beta R)^2 > 1$. The weighted complete-orbit ladder obeys

$$\mathcal{L}_k^{\text{orb}} := \sum_{m=m_\star}^{k-2} G_m R^{2m} = \frac{x^{k-1}}{C_M(x-1)} \{1 + o(1)\}.$$

The absolute radial correction over the whole ladder is only $O(k^{-1})\mathcal{L}_k^{\text{orb}}$, without any sign assumption. Consequently the absolute combined deterministic demand has mass $\mathcal{L}_k^{\text{orb}}\{1 + o(1)\} \rightarrow \infty$.

An exact reverse-triangle argument then shows that vanishing of the direct punctured lower-even subprefix would force the actual signed remainder-plus-parity supply to carry asymptotically at least this full divergent mass. The repository has no moving-order estimate for that supply, so neither closure nor nonclosure follows. Odd orders, upper off-alias orders, determinant gluing, and Gates A–E remain open. No Riemann-hypothesis conclusion is made.

1 The punctured lower-even coefficient family

Work on the same physical clock

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad R = \frac{7}{5}, \quad (1)$$

and choose once and for all an integer $m_\star$ beyond the eventual multiplier-sign threshold in RH-17. For sufficiently large k , put

$$I_k = \{m_\star, m_\star + 1, \dots, k - 2\}. \quad (2)$$

Every $m \in I_k$ gives an even order $n = 2m < 2k - 2$. Thus these orders exclude the critical order $2k$ and the first lower sideband $2k - 2$, and they belong to every strict one-alias prefix $2k < h_\sigma \leq 4k$ [4, 7].

For each $m \in I_k$, let p_{2m} be the primitive period- $2m$ boundary point, and define

$$\Gamma_m = \{|f^j(p_{2m})| : 0 \leq j < 2m\}, \quad M_m = (f^{2m})'(p_{2m}). \quad (3)$$

RH-17 gives $2m$ distinct folded marked points and

$$M_m = -C_M \lambda^m \{1 + o(1)\}, \quad C_M > 0 \quad (4)$$

[1]. Hence, after increasing m_* if needed,

$$G_m = \frac{r_H^{-2m}}{1 + |M_m|} > 0, \quad F_m^{\text{orb}} = 2mG_m. \quad (5)$$

The all-order physical observation identity and deterministic numerator anchor are already fixed in RH-326 and RH-334 [3, 5]. At every non-alias even order $2m < 2k$, the radial counterloop term is exactly

$$\mathcal{A}_{k,2m} = 2(\beta^{2m} - \beta_k^{2m}), \quad \beta_k = \frac{|M_k|^{-1/(2k)}}{r_H}, \quad \beta = \frac{1}{r_H \sqrt{\lambda}}. \quad (6)$$

No sign for (6) is assumed.

Remove the finite set Γ_m from the raw trace partition at order $2m$, and call the resulting quantity $\mathcal{T}_{k,m}^{\text{rest}}$. A finite set has zero localized noisy trace, while each deterministic marked point has weight $(1 + |M_m|)^{-1}$.

Theorem 1.1 (Simultaneous complete-orbit ladder). *For every sufficiently large k and every $m \in I_k$,*

$$\mathcal{T}_{\sigma,2m} = \mathcal{T}_{k,m}^{\text{rest}} - F_m^{\text{orb}}, \quad (7)$$

$$q_{\sigma,k,2m} = \mathcal{T}_{k,m}^{\text{rest}} + \mathcal{P}_{\sigma,2m} - \mathcal{A}_{k,2m} - F_m^{\text{orb}}, \quad (8)$$

$$p_{\sigma,k,2m} = Y_{k,m} + \mathcal{P}_{\sigma,2m} - S_{k,m}, \quad (9)$$

where

$$Y_{k,m} = \mathcal{T}_{k,m}^{\text{rest}} - d_{\sigma,k,2m}, \quad S_{k,m} = F_m^{\text{orb}} + \mathcal{A}_{k,2m}. \quad (10)$$

The coefficient q is the Hardy full-trace constituent, whereas $p = q - d$ is the direct coefficient.

Proof. The proof of the complete extraction in RH-346 depends only on the period- $2m$ orbit, the folding bijection, and the zero noisy trace of a finite set, not on the special relation $m = k - 1$ [2]. It therefore gives (7) for every $m \in I_k$. Substitute (7) into the all-order five-slot identity to obtain (8); then use $p = q - d$ and (10). $\square$

This is a simultaneous coefficient family, not a sum of orbit weights taken from different data types. The same physical k and the same deterministic numerator anchor are retained at every order.

2 Exact weighted orbit ladder

Give each coefficient its prefix weight

$$w_m = \frac{R^{2m}}{2m}. \quad (11)$$

By (5), the complete atom has the exact weighted identity

$$F_m^{\text{orb}} w_m = G_m R^{2m}. \quad (12)$$

Define

$$\mathcal{L}_k^{\text{orb}} = \sum_{m \in I_k} F_m^{\text{orb}} w_m = \sum_{m=m_*}^{k-2} G_m R^{2m}, \quad x = (\beta R)^2. \quad (13)$$

The strict inequality $x > 1$ is source-certified by RH-336 [6].

Theorem 2.1 (Sharp punctured orbit-ladder mass). *As $k \rightarrow \infty$,*

$$\mathcal{L}_k^{\text{orb}} = \frac{x^{k-1}}{C_M(x-1)} \{1 + o(1)\}. \quad (14)$$

In particular, $\mathcal{L}_k^{\text{orb}} \rightarrow \infty$ exponentially.

Proof. Equations (4) and (5) give

$$G_m R^{2m} = \frac{1}{C_M} (\beta R)^{2m} \{1 + o(1)\} = \frac{1}{C_M} x^m \{1 + o(1)\}.$$

For a sequence asymptotic to $C_M^{-1} x^m$ with $x > 1$, summing from a fixed lower endpoint through $k-2$ preserves the geometric-tail asymptotic. More explicitly, split the sum at a fixed large M , bound the relative error by an arbitrary ε on $m \geq M$, and note that the finite initial sum is $o(x^{k-1})$. Since

$$\sum_{m=M}^{k-2} x^m = \frac{x^{k-1} - x^M}{x-1},$$

letting $\varepsilon \downarrow 0$ proves (14). $\square$

Thus the two previously selected orders do not exhaust the deterministic boundary-orbit demand. Even after deleting them, the remaining lower-even orbit ladder has an explicit divergent weighted mass.

3 The radial ladder is aggregate lower order

The finite counterloop radius satisfies

$$\beta_k = \beta \exp \left[-\frac{\log C_M}{2k} + o(k^{-1}) \right]. \quad (15)$$

Set $\varepsilon_k = \log(\beta_k/\beta)$. Then $|\varepsilon_k| \leq C/k$ for all large k . Uniformly for $m \leq k$,

$$|e^{2m\varepsilon_k} - 1| \leq 2m|\varepsilon_k| e^{2m|\varepsilon_k|} \leq C_1 \frac{m}{k}. \quad (16)$$

Using (6), define the absolute radial mass

$$\mathcal{L}_k^{\text{rad,abs}} = \sum_{m \in I_k} |\mathcal{A}_{k,2m}| w_m. \quad (17)$$

Theorem 3.1 (Aggregate radial suppression). *There is a constant C_2 such that*

$$\mathcal{L}_k^{\text{rad,abs}} \leq \frac{C_2}{k} \sum_{m=m_*}^{k-2} x^m = O(k^{-1}) \mathcal{L}_k^{\text{orb}}. \quad (18)$$

Consequently, with

$$\mathcal{L}_k^S = \sum_{m \in I_k} S_{k,m} w_m, \quad \mathcal{L}_k^{S,\text{abs}} = \sum_{m \in I_k} |S_{k,m}| w_m, \quad (19)$$

one has

$$\mathcal{L}_k^S = \mathcal{L}_k^{\text{orb}} \{1 + O(k^{-1})\}, \quad \boxed{\mathcal{L}_k^{S,\text{abs}} = \mathcal{L}_k^{\text{orb}} \{1 + O(k^{-1})\}}. \quad (20)$$

In particular, both are eventually positive and diverge with the asymptotic in (14).

Proof. Equations (6) and (16) imply

$$|\mathcal{A}_{k,2m}| = 2\beta^{2m}|1 - e^{2m\varepsilon_k}| \leq C_3 \frac{m}{k} \beta^{2m}.$$

After multiplication by w_m , this is at most $C_4 x^m/k$. Summation and theorem 2.1 give (18). Finally,

$$|\mathcal{L}_k^S - \mathcal{L}_k^{\text{orb}}| \leq \mathcal{L}_k^{\text{rad,abs}}, \quad |\mathcal{L}_k^{S,\text{abs}} - \mathcal{L}_k^{\text{orb}}| \leq \mathcal{L}_k^{\text{rad,abs}},$$

which proves (20). $\square$

The theorem uses an absolute estimate only to compare two deterministic demand ledgers. It does not use separate absolute bounds to decide the signed physical residual.

4 Necessary aggregate compensation

For the direct coefficient, collect the actual signed supply as

$$Z_{k,m} = Y_{k,m} + \mathcal{P}_{\sigma,2m}. \quad (21)$$

Then (9) reads $p_{\sigma,k,2m} = Z_{k,m} - S_{k,m}$. Define the punctured lower-even residual and supply masses

$$\mathcal{E}_k^{\text{low}} = \sum_{m \in I_k} |p_{\sigma,k,2m}| w_m, \quad (22)$$

$$\mathcal{C}_k^{\text{low}} = \sum_{m \in I_k} |Z_{k,m}| w_m. \quad (23)$$

Both are formed in the direct coefficient data type on the common physical clock.

Theorem 4.1 (Punctured ladder compensation demand). *For every sufficiently large k ,*

$$\boxed{\mathcal{C}_k^{\text{low}} + \mathcal{E}_k^{\text{low}} \geq \mathcal{L}_k^{S,\text{abs}}}. \quad (24)$$

Consequently, if

$$\mathcal{E}_k^{\text{low}} \longrightarrow 0, \quad (25)$$

then

$$\liminf_{k \rightarrow \infty} \frac{\mathcal{C}_k^{\text{low}}}{\mathcal{L}_k^{\text{orb}}} \geq 1, \quad \mathcal{C}_k^{\text{low}} \longrightarrow \infty. \quad (26)$$

Thus vanishing of the punctured lower-even direct subprefix requires actual signed supply with the full leading orbit-ladder mass.

Proof. For each $m \in I_k$, the reverse triangle inequality gives

$$|Z_{k,m}| + |Z_{k,m} - S_{k,m}| \geq |S_{k,m}|.$$

Multiply by w_m and sum to obtain (24). If (25) holds, then $\mathcal{E}_k^{\text{low}}/\mathcal{L}_k^{\text{orb}} \rightarrow 0$ because $\mathcal{L}_k^{\text{orb}} \rightarrow \infty$. Divide (24) by $\mathcal{L}_k^{\text{orb}}$ and use (20). $\square$

The full-trace analogue omits $d_{\sigma,k,2m}$ from $Y_{k,m}$ and is proved identically. Neither version supplies an upper bound for the actual signed supply. A divergent deterministic demand may, in principle, be cancelled coefficient by coefficient.

Remark 4.2 (Why this does not close E_{off}). The lower-even ladder is only one subfamily of the punctured prefix. Odd orders and orders above the first alias are absent. More importantly, theorem 4.1 is a necessary mass law, not a proof that the actual $Z_{k,m}$ fails to meet it. The repository contains no moving-order signed theorem for $\mathcal{T}_{k,m}^{\text{rest}}$, $d_{\sigma,k,2m}$, and $\mathcal{P}_{\sigma,2m}$ jointly. Hence both lower-even closure and lower-even nonclosure remain open, as does the full E_{off} aggregate.

5 Executable protocol and next route

The artifact reconstructs the complete boundary multipliers and exact finite ladder at $k = 8, 12, 16, 20$. It checks the exact identity $F_m^{\text{orb}} w_m = G_m R^{2m}$, the geometric orbit reference, the absolute radial and combined-demand ratios, and a rational reverse-triangle fixture. All finite rows are formula-reproduction checks only. They are not interval certificates, asymptotic evidence, or observations of an actual noisy operator.

RH-348 proves an exact punctured lower-even physical coefficient ladder, its sharp aggregate deterministic demand, and a necessary aggregate compensation mass law. It proves no signed supply estimate and no prefix closure or nonclosure.

RH-349 may now compare the first two still-punctured sidebands $m = k - 2$ and $m = k - 3$. Their scalar parity/demand ratios differ by the fixed factor λ , so the minimax identity

$$\inf_{a>0} \max\{|a - 1|, |a/\lambda - 1|\} = \frac{\lambda - 1}{\lambda + 1} > 0$$

can yield a conditional two-order obstruction under explicit target-negligibility hypotheses for both actual signed remainders. The missing hypotheses may not be inferred from the present demand theorem.

RH-288 remains inactive. Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt prime-power trace, proves no completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

References

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